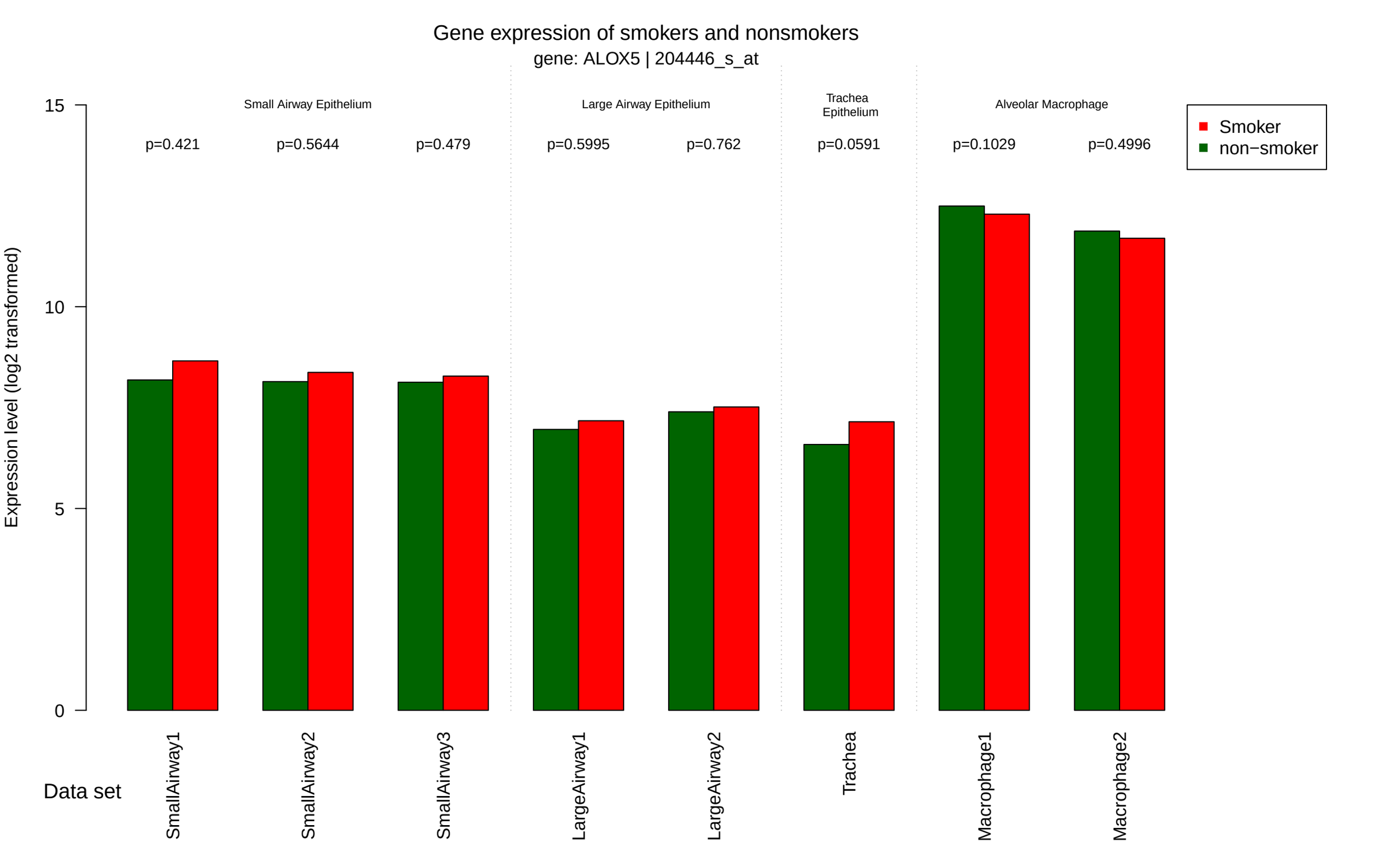
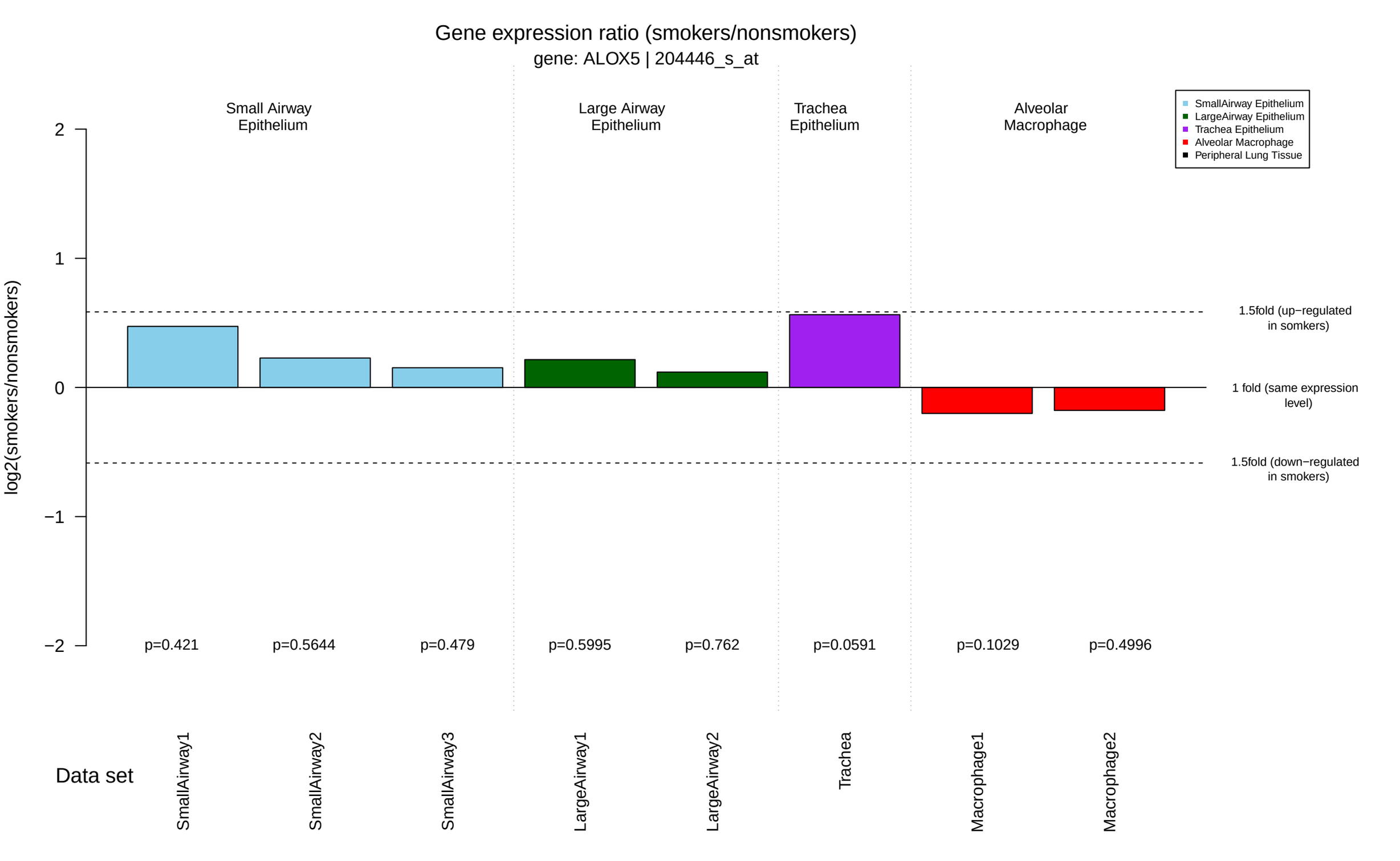
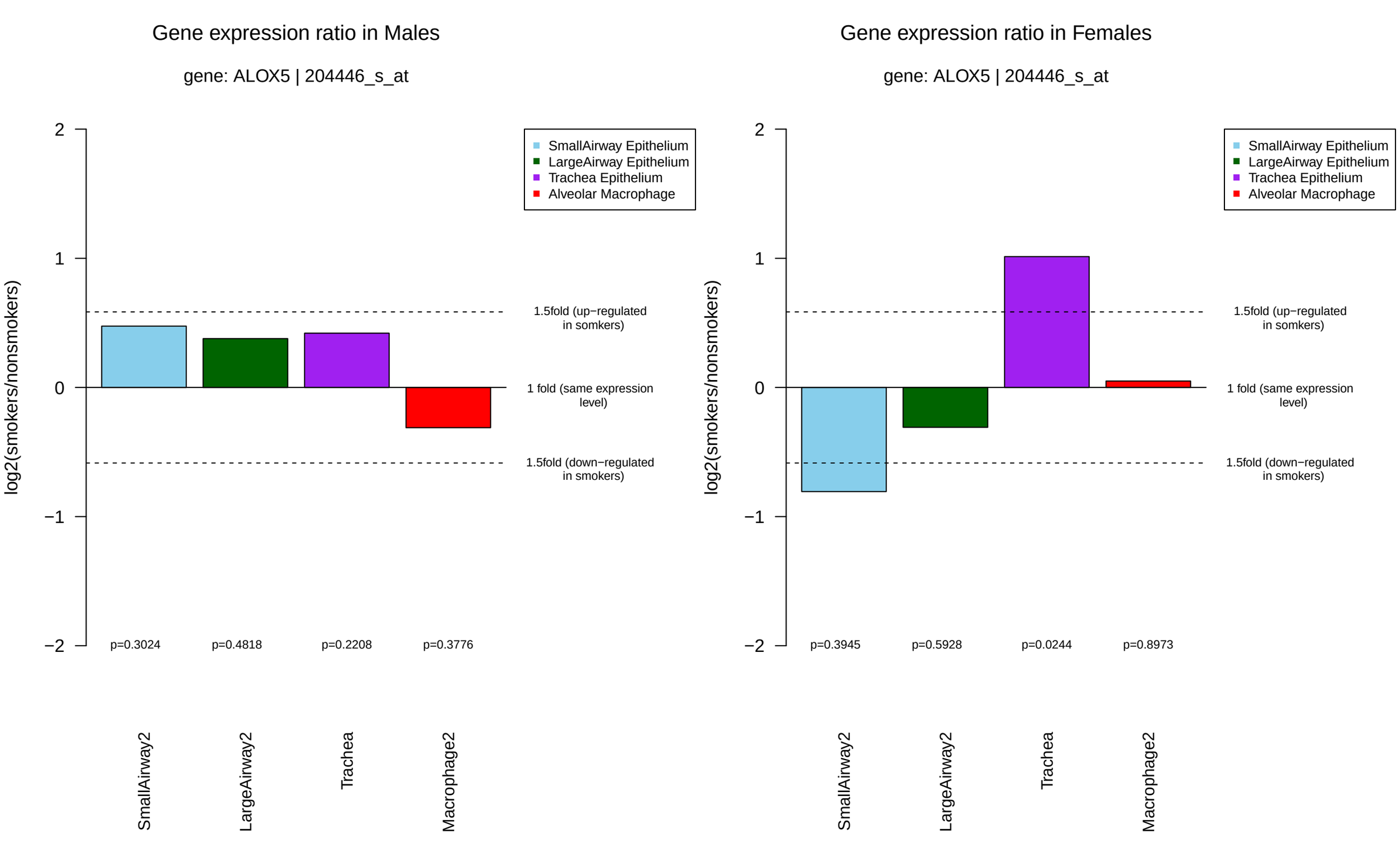
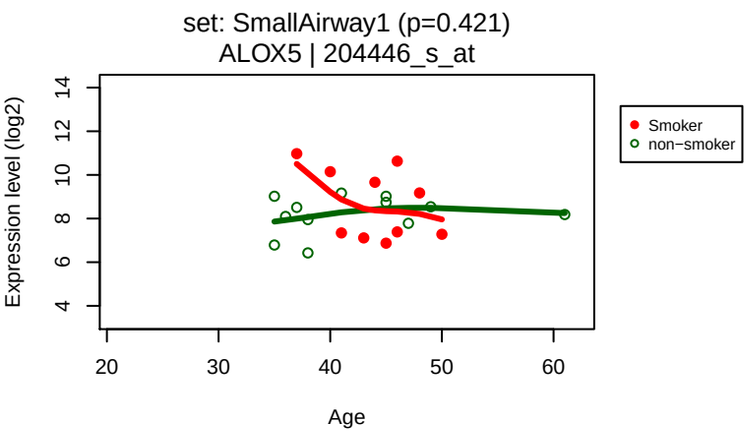
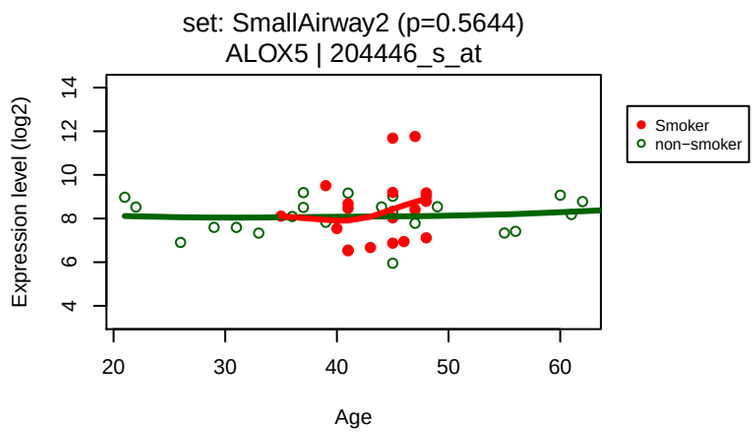
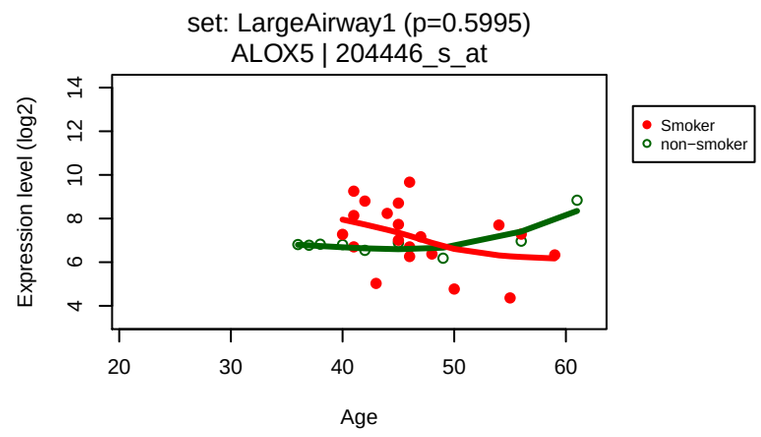

Gene expression of smokers and nonsmokers
gene: ALOX5 | 204446_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.421
p=0.5644
p=0.479
p=0.5995
p=0.762
p=0.0591
p=0.1029
p=0.4996
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

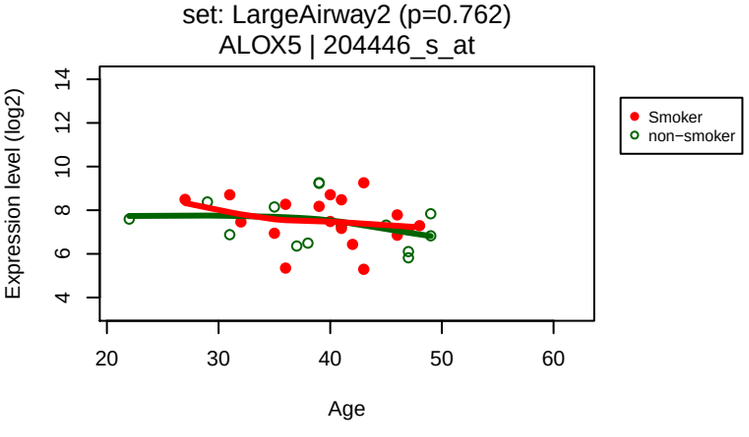

Gene expression ratio (smokers/nonsmokers)
gene: ALOX5 | 204446_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.421
p=0.5644
p=0.479
p=0.5995
p=0.762
p=0.0591
p=0.1029
p=0.4996
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

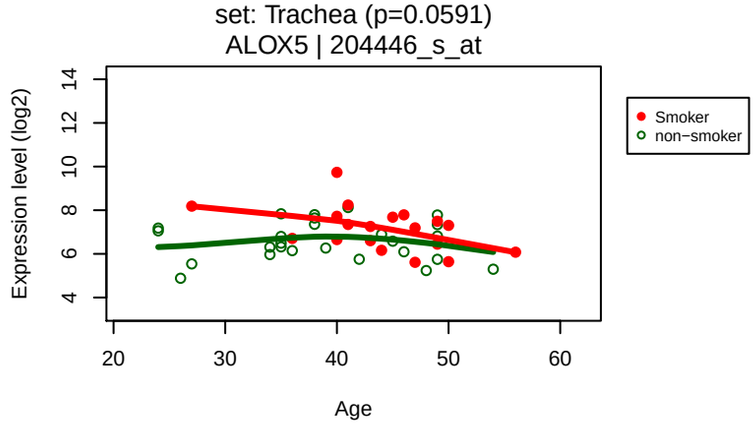

Gene expression ratio in Males
gene: ALOX5 | 204446_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3024
p=0.4818
p=0.2208
p=0.3776
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALOX5 | 204446_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3945
p=0.5928
p=0.0244
p=0.8973
SmallAirway2
LargeAirway2
Trachea
Macrophage2

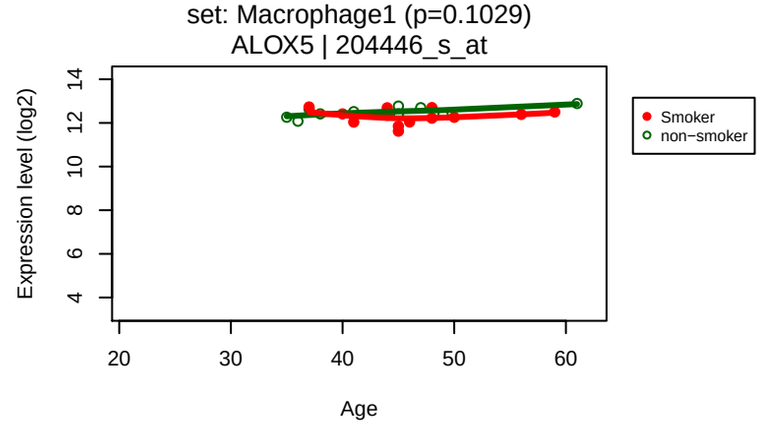

set: SmallAirway1 (p=0.421)
ALOX5 | 204446_s_at
Expression level (log2)
Age
Smoker
non−smoker

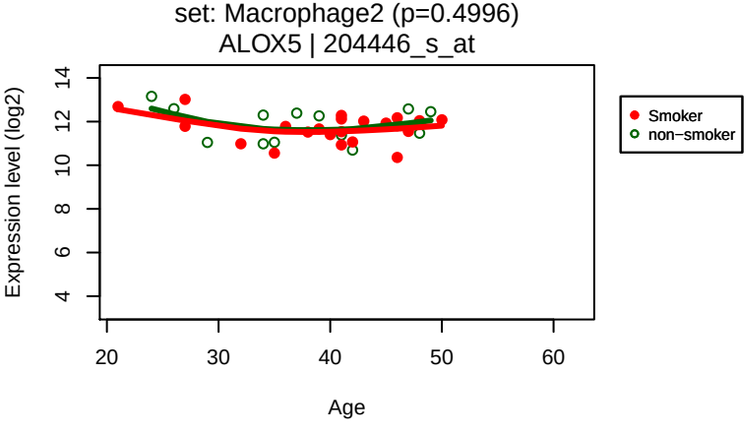

set: SmallAirway2 (p=0.5644)
ALOX5 | 204446_s_at
Expression level (log2)
Age
Smoker
non−smoker

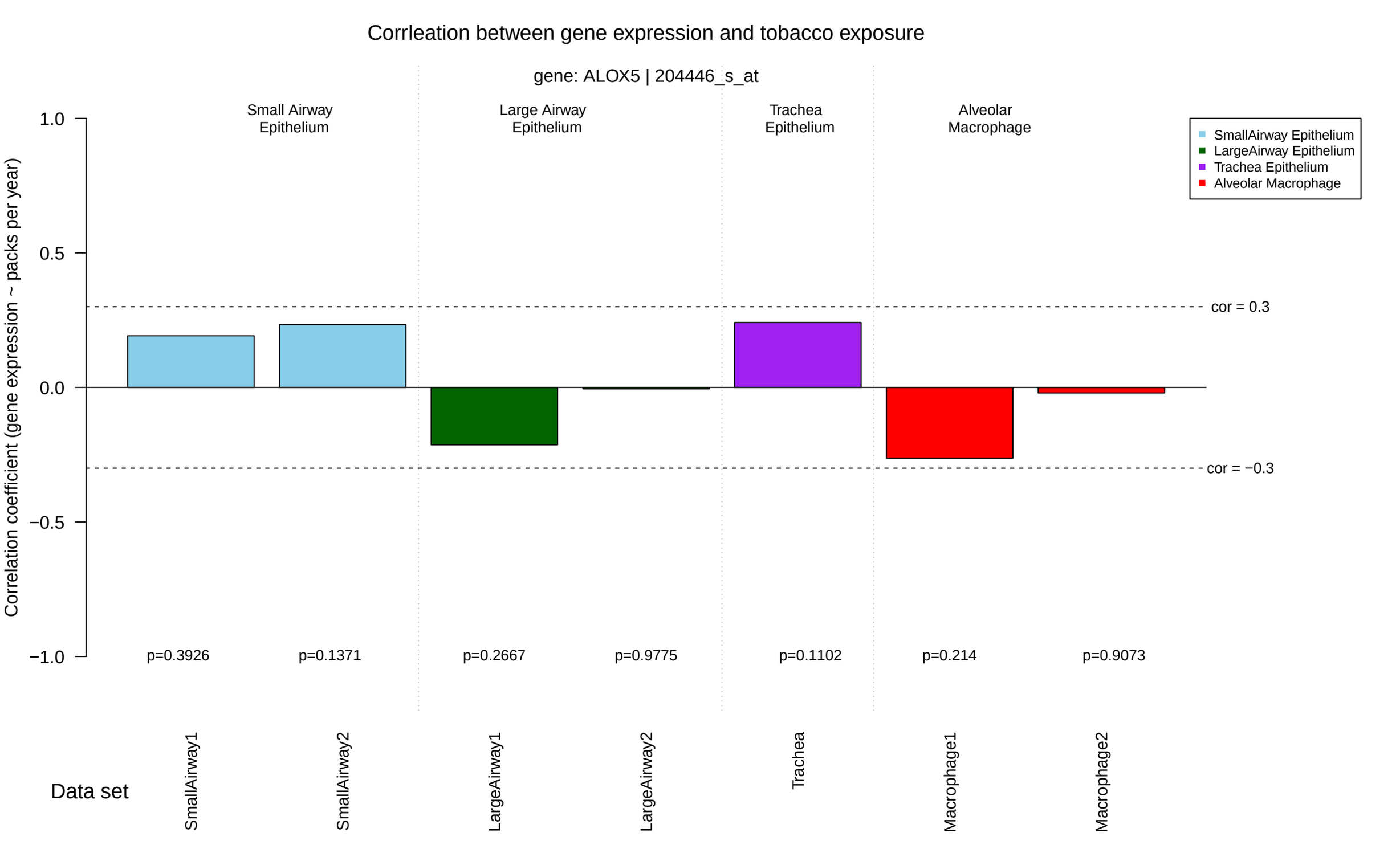

set: LargeAirway1 (p=0.5995)
ALOX5 | 204446_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.762)
ALOX5 | 204446_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0591)
ALOX5 | 204446_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1029)
ALOX5 | 204446_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4996)
ALOX5 | 204446_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALOX5 | 204446_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3926
p=0.1371
p=0.2667
p=0.9775
p=0.1102
p=0.214
p=0.9073
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2